

Double Bunch Self-Seeding at LCLS-II

A. Halavanau

SLAC National Accelerator Laboratory

September 2, 2019



Reaching high power in XFEL

Why do we want to increase XFEL power?

As Stanford University's Philip Bucksbaum points out, the capability exists to raise laser intensities another nine orders of magnitude. That milestone will be accomplished, he predicts, using a CPA-based petawatt laser in tandem with an electron beam such as the one at SLAC or at DESY in Hamburg, Germany. When laser pulses are slammed into the electron beam, their intensity in the electron's frame of reference will be boosted by a value of $4\gamma^2$, where γ is the relativistic Lorentz factor, $(1-v^2/c^2)^{-1/2}$, with v the

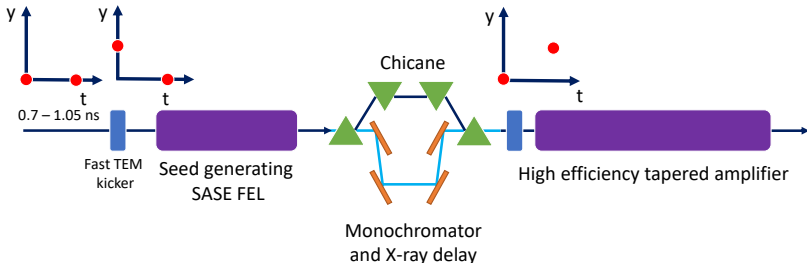
Physics Today (12/2018)

- Extending the photon energy range with harmonic generation
- New high-field atomic physics
- Single particle imaging and inelastic X-ray scattering
- Back-scattering of XFEL pulse with electrons/photons (new QED colliders)

Previous considerations:

- E. L. Saldin, E. A. Schneidmiller, Yu. V. Shvyd'ko, and M. V. Yurkov. AIP conf., 581(1), pp. 153–161, (2001).
- G. Geloni *et al.*, pulse doubler (2010), arXiv:1007.2743
- Y. Ding, Z. Huang, R. Ruth, (2010), PRSTAB 13, 6,
- C. Emma, *et al*, Appl. Phys. Lett., 110(15):154101, (2017)

Double bunch FEL schematics



- First bunch lases in the first undulator; second bunch is off-set
- SASE radiation overlaps with the second bunch after monochromator and chicane delay lines
- Second bunch is “fresh”, very high seed power from first

Consider minimum LCLS-II hardware changes, what is the maximum X-ray power we can get with this setup?

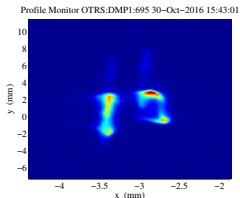
Double bunch mode of LCLS

- Two laser pulses arrive with a delay on photocathode
- Setup can generate 0.35 - 250 ns separation
- Possible to control charge, energy, energy spread of two bunches

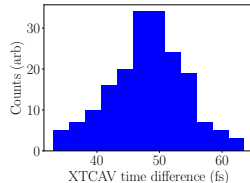
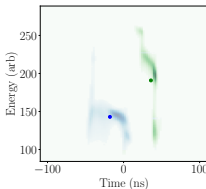


Add second laser to get larger separation and more flexibility.

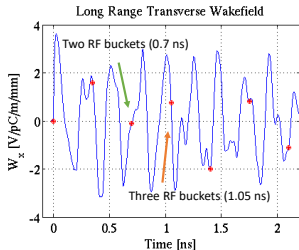
$$\sigma_{\tau_{L_1 L_2}}^2 = \sigma_{\tau_{L_1}}^2 + \sigma_{\tau_{L_2}}^2 \quad (\text{more jitter, but more control; measured jitter is 6 fs at 50 ns } \Delta\tau)$$



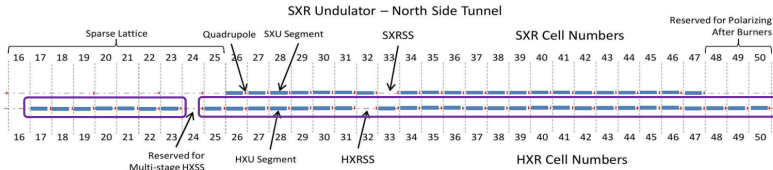
Demonstrated experimentally by F. J. Decker (2010)



Linac wakefield effects



- $W_y(t) = \sum 2\kappa_{y1n} \sin(2\pi f_{1n}t) e^{-\pi f_{1n}t/Q_{1n}}$
- Transverse wakefields in the CuRF linac define bunch separation
- Example: 0.1 mm offset in first bunch results in 20 μm for second bunch at 1.05 ns sep. (no effect at 0.7 ns).



CuRF is followed by soft X-ray (SXR) and hard X-ray (HXR) undulators

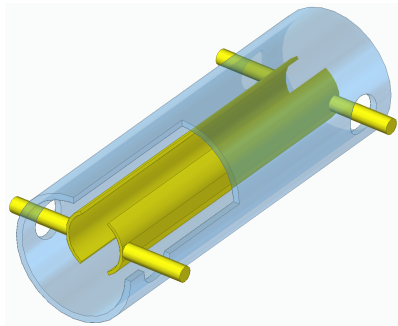
Design study and TEM kicker

Initial assessment:

- ① Two bunches have identical charge, energy spread, emittance
- ② Two bunches have well defined controlled orbits
- ③ Start-to-end simulations for electron beam tracking
- ④ Crystals are considered to be diffraction limited (perfect)

Ultra-fast kicker:

- ① Kicker strength $\alpha = \frac{2VL}{r} \frac{4}{\pi} \sin \frac{\theta}{2}$,
 $\theta = \pi/3$
- ② Kick angle θ :
 $L_g = L_{g0}/(1 - \theta^2/\theta_C^2)$, need
 $\theta = \theta_C = \sqrt{\lambda/L_{g0}}$.
- ③ Required peak voltage 10 kV
- ④ Stripline length 0.2 m



Beam parameters in simulations

Electron beam energy	6.5 - 9 GeV
Peak Current	4 kA
Pulse duration	15 fs
Normalized emittance	0.4 μm
Energy spread at input of tapering section	2.5 MeV
Beta function	10 m
Photon energy	4 keV – 8 keV
Gain length	~ 2 m

$$P_{coh}(z) = \frac{Z_0 K^2 J J(z)^2 I_{pk}^2 b^2 z^2}{32 \sqrt{2} \pi \sigma^2 \gamma^2}$$

We selected conservative beam parameters, based on LCLS copper linac operations; notice strongly dependent on σ^2 (focusing)

Undulator tapering

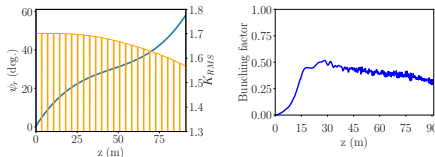
In this approach we assign a variation of the resonant phase and evaluate the undulator parameter from:

$$\frac{dK}{dz} = -\frac{e}{mc^2} \frac{2\lambda}{\lambda_U} JJ(z)E(z) \sin \psi_r$$

$$K_{n+1} = K_n + \alpha_n E_n \sin \psi_{r,n}$$

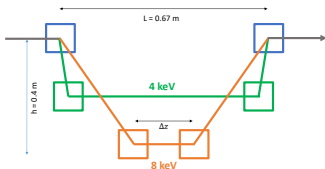
We use step-wise evaluation of the tapered section, every 5 periods, calculating the radiation electric field from Genesis. [Strategy for the resonant phase variation](#) :

- Linear growth in first few sections
- Slow growth in the middle
- Rapid growth towards the end



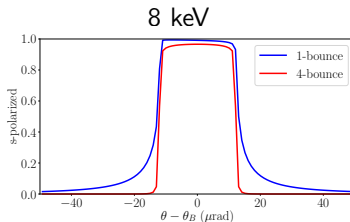
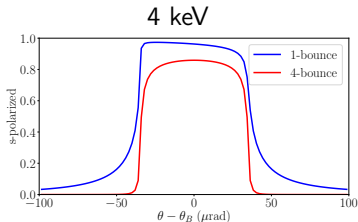
Still room for improvement, e.g. considering zig-zag tapering

4 crystal monochromator



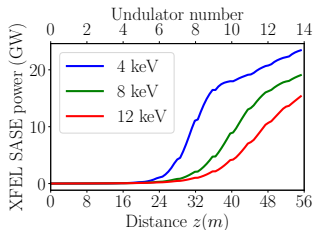
- $\Delta_z = L - c\Delta\tau(\cot^2 \theta - 1)/2$
- $h = c\Delta\tau/2 \tan \theta$
- $\Delta\omega/\omega = \Delta\theta/\tan \theta$

- 1 C*(111) crystals will be used as mirrors
- 2 Monochromator dimensions: 0.9 m x 0.5 m
- 3 Beam stopper in the middle to block SASE

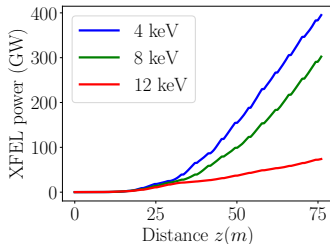
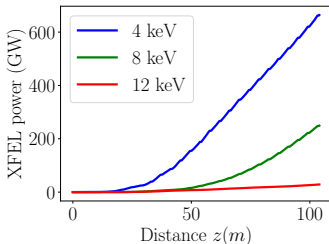


No “pre-pulse” compared to existing self-seeding

Tapered section using HXR

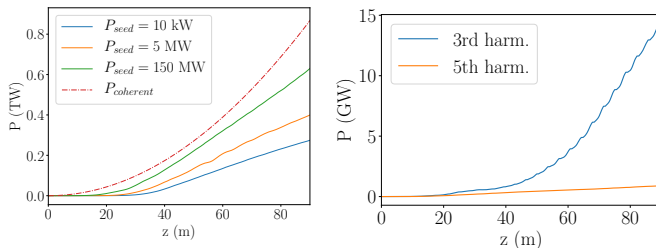


- Initial SASE simulations define the number of undulators in the first section
- Seeding at U7 (bottom left) and U 14 (bottom right)

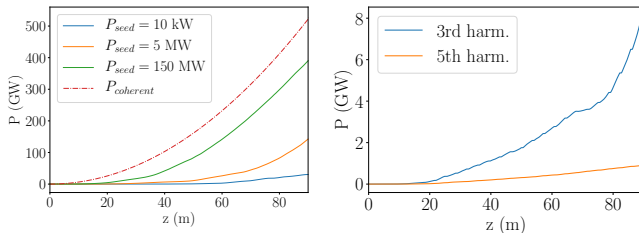


Comparison with self-seeding

Genesis calculation for 4 keV photons (time-dependent)

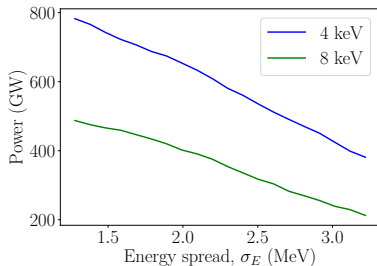
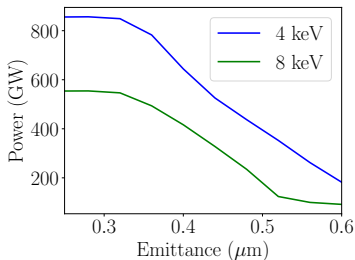


Genesis calculation for 8 keV photons (time-dependent)



Comparison to fresh slice

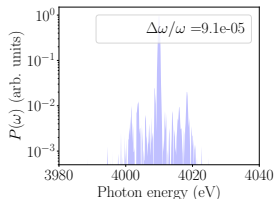
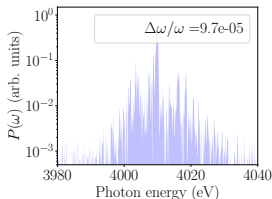
In fresh slice, only about 1/3 of the bunch is used to generate the SASE signal and another 1/3 for the amplification process. Therefore to keep the same amount of charge for lasing, one has to triple the charge, hence increase ϵ , σ .



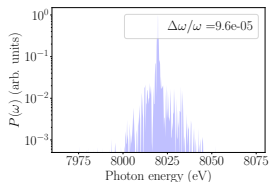
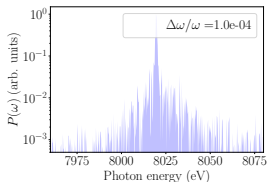
- 1 Double bunch exceeds the performance of fresh slice
- 2 Beam quality is very important

Brightness improvement

Genesis calculation for 4 keV photons (time-dependent)
5 MW seed 82%, in fund. || 150 MW seed, 92% in fund.



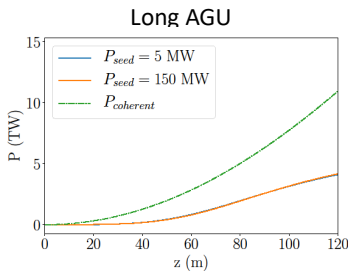
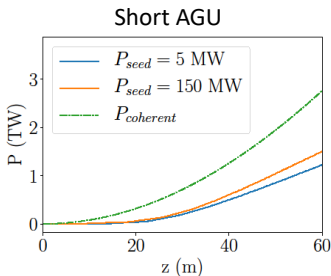
Genesis calculation for 8 keV photons (time-dependent)
5 MW seed 88%, in fund. || 150 MW seed, 96% in fund.



Tapered AGU (superconducting)

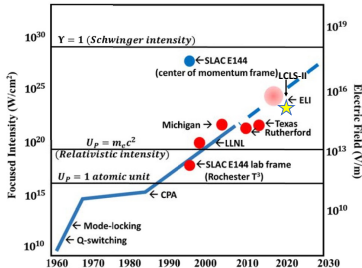
Advanced Gradient Undulator (AGU), if built, can reach multi-TW power for 8 - 12 keV photons, and harmonics if built planar

Calculation for 8 keV



Shorter period, stronger K (superconducting) means strong focusing, shorter drift sections, smaller β -functions is similar to improving beam quality, therefore much better performance. Similarly, at 12 keV 2 TW and 40 GW at 36 keV.

Critical fields with XFEL



Some applications:

- Single particle imaging
- Schwinger field limit (10^{18} V/m if focused to 10 nm)
- Higher harmonics can be used as well (see paper by C. Emma (2016))

- X-ray beamline to back-scatter photons
- R&D in X-ray focusing (how to get 10 nm at high power?)
- QED with strong radiation corrections
- $E = 1.2 \cdot 10^{15} \frac{\sqrt{P/P_0}}{(\sigma/\sigma_0)}$, where $P_0 = 1 \text{ TW}$ and $\sigma_0 = 10 \text{ nm}$.
- When focused to 100 nm $E/E_{cr} \approx 1$, 10 nm $\rightarrow E/E_{cr} \approx 5$

Double-bunch LCLS-II summary

- ① We demonstrated, via numerical simulations, LCLS-II can provide very high power and high-brightness X-rays in the range of 4 keV to 8 keV photons (650 GW and 400 GW respectively)
- ② We could potentially get up to 8 GW of 24 keV photons
- ③ List of additional components include: four crystals monochromator, one ultra-fast kicker
- ④ In order to achieve better performance at 8 keV we recommend shifting HXRSS chicane from U7 to U8.
- ⑤ **Strong focusing undulator provides much higher output power and photon energy range**
- ⑥ More details in Halavanau, *et al*, JSR, **26**, 3 (2019)

On behalf of:

F.-J. Decker, C. Emma, A. Krasnykh, G. Marcus, A.A. Lutman, Z. Huang, Y. Ding, D. Zhu, J. Krzywinski, A. Ratti, A. Marinelli, C. Pellegrini (SLAC)

Thank you for your attention!

